

OpenCPU SDK

SMS API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Wang Chen

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the SMS operation-related APIs in OpenCPU SDK, covering the following aspects:

- Initialization of the SMS module.
- Configuring and retrieving SMS reporting methods and storage locations.
- Retrieving and setting the SMS center number.
- Reading and deleting SMS messages.
- Sending and receiving SMS messages.

2 Data Structure

2.1 typedef enum

```
typedef enum
{
    Nwy_SMS_Msg_Format_GSM7 = 1,
    Nwy_SMS_Msg_Format_8BIT = 2,
    Nwy_SMS_Msg_Format_UCS2 = 3,
}nwy_sms_msg_format_type_e;

typedef enum
{
    Nwy_SMS_PP_IND = 0, // PP msg
    Nwy_SMS_CB_IND = 1, // CB msg
    Nwy_SMS_CB_CMAS_IND = 2, // CB CMAS msg
    Nwy_SMS_SEND_IND = 3, // CB CMAS msg
}nwy_mt_sms_event_e;

typedef enum
{
    Nwy_SMS_Transfer_Unknown = -1,
    Nwy_SMS_Transfer_And_Store,
    Nwy_SMS_Transfer_Only,
} nwy_sms_transfer_type_e;

typedef enum
{
    Nwy_SMS_Transfer_Online_Unknown = -1,
    Nwy_SMS_Transfer_Online_Discard = 1,
    Nwy_SMS_Transfer_Online_Cache,
    Nwy_SMS_Transfer_Online_Direct,
} nwy_sms_transfer_online_mode_e;

typedef enum
{
    Nwy_SMS_Storage_Type_UIM,
    Nwy_SMS_Storage_Type_NV,
    Nwy_SMS_Storage_Type_None = -1,
}nwy_sms_storage_type_e;

typedef enum
{
    Nwy_SMS_Msg_Dflag_None = 0,
    Nwy_SMS_Msg_Dflag_Read = 1,
    Nwy_SMS_Msg_Dflag_Read_Send = 2,
    Nwy_SMS_Msg_Dflag_Read_Send_Unsend = 3,
    Nwy_SMS_Msg_Dflag_All = 4,
    Nwy_SMS_Msg_Dflag_Max,
}nwy_sms_msg_dflag_e;
```

2.2 typedef struct

```
typedef struct nwy_sms_info_type{
    char phone_num[NWY_SMS_MAX_ADDR_LENGTH];
    uint32_t msg_context_len;
    char msg_context[NWY_SMS_MAX_MO_MSG_LENGTH + 1];
    nwy_sms_msg_format_type_e msg_format;
}nwy_sms_info_type_t;

typedef struct nwy_sms_time_stamp
{
    uint16_t uYear;
    uint8_t uMonth;
    uint8_t uDay;
    uint8_t uHour;
    uint8_t uMinute;
    uint8_t uSecond;
    int8_t iZone;
} nwy_sms_time_stamp_t;

typedef struct nwy_sms_rcv_info_type{
    uint16_t nIndex;
    nwy_sms_msg_format_type_e msg_format; // Message format.
    uint32_t msg_content_len;
    uint8_t msg_pdu[NWY_SMS_MAX_PDU_LENGTH + 1];
    uint8_t msg_content[NWY_SMS_MAX_MT_MSG_LENGTH + 1];
    char source_phone_num[NWY_SMS_MAX_ADDR_LENGTH + 1]; // Source address.
    nwy_sms_time_stamp_t date;
    nwy_sms_context_decode_type_e context_decode_type;
    uint8_t msg_decoded_content[NWY_SMS_MAX_MT_MSG_LENGTH + 1];
    nwy_sms_transfer_type_e transfer_type;
    nwy_sms_storage_type_e storage_type;
    nwy_sim_id_e dst_id;
} nwy_sms_rcv_info_type_t; /* Message */

typedef struct {
    uint8_t phone_num[NWY_SMS_MAX_ADDR_LENGTH];
    uint8_t result;
    nwy_sim_id_e dst_id;
} nwy_sms_send_ind_t;

typedef struct {
    char sca[NWY_SMS_MAX_ADDR_LENGTH + 1];
    unsigned char tosca;
} nwy_sms_sca_t;

typedef struct nwy_sms_report_para{
    nwy_sms_transfer_type_e transfer_type;
    nwy_sms_transfer_online_mode_e transfer_online_mode;
} nwy_sms_report_para_t;

typedef struct {
    uint8_t len;
    uint8_t indices[NWY_MAX_MSG_LIST_LEN];
}nwy_sms_msg_list_t;

typedef uint32_t sms_client_handle_t;
```

```
typedef void (*nwy_sms_evt_handler)
(
    nwy_mt_sms_event_e ind_type,
    void *ind_struct
);
```


3 APIs

This interface is defined in `nwy_sms_api.h` and is used to implement SMS sending, receiving, and storage location configuration functionalities.

3.1 `nwy_sms_option_init`

Format	<code>nwy_error_e nwy_sms_option_init(nwy_sim_id_e sim_id)</code>
Description	Initializes SMS parameters.
Parameters	<code>nwy_sim_id_e sim_id</code> : Specifies the SIM card ID to operate.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: another enumeration value

3.2 `nwy_sms_storage_set`

Format	<code>nwy_error_e nwy_sms_storage_set(nwy_sim_id_e sim_id, nwy_sms_storage_type_e sms_storage)</code>
Description	Sets the SMS storage location.
Parameters	<code>nwy_sim_id_e sim_id</code> : Specifies the SIM card ID to operate. <code>nwy_sms_storage_type_e sms_storage</code> 1: Device-side storage. 2: SIM card storage.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: another enumeration value

3.3 `nwy_sms_storage_get`

Format	<code>nwy_error_e nwy_sms_storage_get(nwy_sim_id_e sim_id, nwy_sms_storage_type_e *sms_storage)</code>
Description	Gets the SMS storage location.
Parameters	<code>nwy_sim_id_e sim_id</code> : Specifies the SIM card ID to operate. <code>nwy_sms_storage_type_e sms_storage</code> 1: Device-side storage. 2: SIM card storage.
Return Value	Success: <code>NWY_SUCCESS</code>

Failure: another enumeration value

3.4 nwy_sms_report_set

Format	nwy_error_e nwy_sms_report_set(nwy_sim_id_e sim_id, nwy_sms_report_para_t report_para)
Description	Sets the SMS reporting method.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sms_report_para_t report_para: Parameters for new SMS reporting.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.5 nwy_sms_sca_set

Format	nwy_error_e nwy_sms_sca_set(nwy_sim_id_e sim_id, nwy_sms_sca_t sca)
Description	Sets the SMS center number.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sms_sca_t sca: SMS center number information.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value
Note	Ensure the SMS center number is correct by confirming with the local operator.

3.6 nwy_sms_sca_get

Format	nwy_error_e nwy_sms_sca_get(nwy_sim_id_e sim_id, nwy_sms_sca_t *sca)
Description	Gets the SMS center number.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sms_sca_t sca: SMS center number information.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.7 nwy_sms_msg_send

Format	nwy_error_e nwy_sms_msg_send(nwy_sim_id_e sim_id, nwy_sms_info_type_t *p_sms_data)
Description	Sends an SMS.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate.

	nwy_sms_info_type_t *p_sms_data: Information about the SMS to be sent.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value
Note	The result only indicates the success of the send action, not whether the SMS was successfully sent over the network.

3.8 nwy_sms_rcv_cb_reg

Format	nwy_error_e nwy_sms_rcv_cb_reg (nwy_sim_id_e sim_id, nwy_sms_evt_handler evt_handler)
Description	Registers a callback function for receiving SMS notifications.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID, allowing separate handling for multiple cards. nwy_sms_evt_handler evt_handler: Pointer to the user-implemented function to process SMS content. typedef void (*nwy_sms_evt_handler) (nwy_sim_id_e sim_id, nwy_mt_sms_event_e urc_type, nwy_sms_info_ind_t *ind_struct);
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.9 nwy_sms_rcv_cb_unreg

Format	nwy_error_e nwy_sms_rcv_cb_unreg (nwy_sim_id_e sim_id, nwy_sms_evt_handler evt_handler)
Description	Unregisters the callback function for receiving SMS notifications.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID, allowing separate handling for multiple cards. nwy_sms_evt_handler evt_handler: Pointer to the user-implemented function to process SMS content. typedef void (*nwy_sms_evt_handler) (nwy_sim_id_e sim_id, nwy_mt_sms_event_e urc_type, nwy_sms_info_ind_t *ind_struct);
Return Value	Success: NWY_SUCCESS

Failure: another enumeration value

3.10 nwy_sms_msg_del

Format	nwy_error_e nwy_sms_msg_del(nwy_sim_id_e sim_id, uint16_t index)
Description	Deletes an SMS.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. uint16_t index: Index of the SMS to delete.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.11 nwy_sms_msg_read

Format	nwy_error_e nwy_sms_msg_read(nwy_sim_id_e sim_id, uint16_t index, nwy_sms_rcv_info_type_t *sms_data);
Description	Reads an SMS.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. uint16_t index: Index of the SMS to read. nwy_sms_rcv_info_type_t *sms_data: SMS content information.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.12 nwy_sms_msg_del_ext

Format	nwy_error_e nwy_sms_msg_del_ext(nwy_sim_id_e sim_id, nwy_sms_msg_dflag_e dflag)
Description	Deletes SMS messages of a specified type.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sms_msg_dflag_e dflag: Specifies the type of SMS to delete.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.13 nwy_sms_msg_list_read

Format	nwy_error_e nwy_sms_msg_list_read(nwy_sim_id_e sim_id, nwy_sms_msg_list_t *sms_list)
Description	Gets the indices and types of all SMS messages.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate.

	nwy_sms_msg_list_t *sms_list: List of SMS messages.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value